Callguard demo testing guide

Run this every time you test at home. Score each run on the sheet at the back. When the score says you're ready, record the submission video. **New in this edition:** the live call runs on **Google Meet** with **AirPods as your mic** and the Mac speakers playing the caller (page 4).

The three demo scenarios

#	Script file	What happens	What we want to see
1 (main pitch)	s1_jess.txt	Jess's call: 'Ministry of Public Security', money-laundering suspect, 'go to a quiet room alone', bank login, 'detained for two years'	Green at lines 1-2, amber at line 3, **red at line 5** (the secrecy line), the pause screen at line 6
2 (backup scam)	s3_it_code.txt	Fake IT: 'read me the code we just sent to your phone'	Green at lines 1-2, amber at line 3, **red at line 4**, the pause screen opens
3 (safe call)	b3_real_bank.txt	Real bank: 'if you don't recognise the charge, hang up and call the number on your card'	Green at lines 1-2, **never red**, no pause screen. Proves no false alarms

A few words you'll see

Word	Plain meaning
Beat	One caller line. After every caller line Callguard re-checks the call and updates the meter.
Meter levels	**low** (green, nothing wrong) → **watch** (amber, be careful) → **danger** (red, full warning).
Giveaway	The line where the scam becomes obvious. The screen must be red by this line, and should not be red before it.
Pause (gate)	The 2-minute 'safe pause' screen that opens when the caller asks for money, a code or a login.
Eval	A command that feeds the script to the local AI model line by line and checks each result automatically. No audio, no window. It's the fastest test.
Fallback	The model didn't answer in time, so Callguard showed a safe default (low). Usually means the laptop or Ollama is slow.
p95 latency	How long the slowest checks take (95% of checks are faster than this). Under 8 s means the warning keeps up with the call.
rms	How loud a 3-second chunk of audio is. Silence about 0.001, speaker echo about 0.003-0.007, your voice into the AirPods about 0.013-0.06.
Mic floor	CALLGUARD_MIC_MIN_RMS: your mic chunks quieter than this are ignored, so the caller's voice leaking from the speakers isn't written down as 'you'.

How scoring works

Each run has three parts. Add them up for a score out of 100.

Part	What you do	Time	Points
A. Eval numbers	Run one command, copy the numbers	~3 min	40
B. On-screen check	Replay each script in the app window and tick what you see	~8 min	30
C. Live call rehearsal	A friend reads Jess's script on Google Meet from another room; you watch and time it	~15 min	30
	Total	~25 min	**100**

Ready to record when: two runs in a row score **85 or more**, AND every run passes the three must-haves: (1) Jess's call is red by line 5 and not before, (2) the safe bank call is never red, (3) no fallback in Part A. A must-have failure means don't record yet, whatever the score.

Before every run (2 min)

1. Plug in the laptop (a slow laptop makes the model slow). Quit **Spotify, Slack, Mail, WhatsApp, Discord** and anything that pops up or plays sound. Do Not Disturb **off**, so the Callguard notification can show.

2. Open Terminal, go to the project and turn on its Python environment (you'll see `(.venv)`):

```
cd ~/Desktop/scam-call-detector
source .venv/bin/activate
```

3. Get the code you're testing. Until the mic-floor change is merged into main, test its branch:

```
git fetch origin claude/charming-faraday-tbyue2
git checkout claude/charming-faraday-tbyue2
git pull
```

Once it's merged: `git checkout main && git pull`. Check which one you're on any time with `git branch --show-current`.

4. Start Ollama (menu-bar app, or `ollama serve`), then warm up the model so the first check isn't slow:

```
ollama run qwen2.5:7b "hi"
```

5. Check the offline tests still pass (should say 'passed', nothing red):

```
pytest tests/detection tests/audio -q
```

Part A: Eval numbers (40 points)

The eval sends each script to the model line by line and marks each line pass or FAIL. There are 17 beats in total: Jess 7, IT 5, bank 5.

1. Run:

```
python -m callguard.detection.eval --script s1_jess --script s3_it_code --script b3_real_bank
```

2. Read the table. Each row is one caller line. The last column shows a tick mark (pass) or FAIL. '(fallback)' next to it means the model timed out.

3. Read the last line, which starts with SUMMARY. It looks like:

```
SUMMARY beats 17/17 pass · scam giveaways danger 2/2 · twins danger 0/5 · twins low 4/5 · p95 latency 7.5 s
```

4. Score it:

Item	Where to find it	Points
Beats passed	SUMMARY 'beats X/17'. Points = X ÷ 17 × 30, rounded down (17 = 30, 16 = 28, 15 = 26, 14 = 24)	0-30
Scam giveaways	'scam giveaways danger 2/2' = 5. Anything less = 0	0 or 5
Speed	p95 under 8 s = 3. 8-12 s = 1. Over 12 s = 0	0-3
No fallback	No '(fallback)' anywhere in the table = 2	0 or 2
	Part A total	**/40**

5. If anything FAILed, run it again with -v to see why. Save that output for Claude:

```
python -m callguard.detection.eval -v --script s1_jess > ~/Desktop/eval-run.txt
```

Must-have: s1_jess beats 1-2 must say low and beat 5 must say danger. 'twins danger' must be 0/5 (the safe bank call never red). Any '(fallback)' = not ready.

Part B: On-screen check (30 points)

Now watch the real app window while a script plays in, as if it were a call. This tests what the judges will see. No mic needed. **Wi-Fi on.**

1. Start the app with Jess's script:

```
python -m callguard.server --fake tests/detection/scripts/s1_jess.txt
```

2. Watch, and tick each box on the score sheet as it happens. When the call finishes, click **Hang up and call back**, then **End call**. Close the window (Ctrl+C in Terminal).
3. Repeat for `s3_it_code.txt` and `b3_real_bank.txt`.
4. **Wi-Fi off**, replay Jess's script one more time. Then turn Wi-Fi back on.

What to tick (2 points each)

#	Scenario	Check	Pts
B1	Jess	Lines 1-2 green. Lines 3-4 amber, not red	2
B2	Jess	Line 5 ('go to a quiet room alone...'): window turns red, tone plays, Mac notification shows	2
B3	Jess	Red screen shows tactic pills (e.g. 'Claims authority', 'Keep it secret') with the caller's exact words when clicked	2
B4	Jess	Line 6 (bank login): the pause screen opens with the countdown	2
B5	Jess	Lookup shows 'Ministry of Public Security'; footer shows only that name left the device	2
B6	Jess	**Hang up and call back** shows the big number; **End call** wipes the panel clean	2
B7	IT	Lines 1-2 green, line 3 amber, line 4 ('read it to me') red	2
B8	IT	The pause screen opens; the rule 'IT will never ask you for a code or password' shows	2
B9	Bank	Lines 1-2 green, then green or amber. **Never red**	2
B10	Bank	No pause screen opens	2
B11	Wi-Fi off	Header says 'Offline'	2
B12	Wi-Fi off	Jess still turns red at line 5	2
B13	Wi-Fi off	Lookup says 'Offline, search skipped'; egress log says 'Offline, not sent'	2
B14	Any	No yellow banner ('Basic checks only...') during any run	2
B15	Any	Nothing looks broken: no overlapping text, cut-off words or frozen meter	2
		Part B total	**/30**

Known quirk, don't deduct: on Jess's line 6 the rule shown may be 'IT will never ask you for a code or password'. That rule fires on any password request. It's fixable with a new rule if you want it.

Live audio setup: Google Meet + AirPods (do before Part C)

How the sound gets in. Callguard tells who is talking by **which input** the sound comes from, not by voice. **Caller** = Meet's sound, sent by the Mac to the speakers *and* to BlackHole at the same time (the Multi-Output Device). **You** = your AirPods mic. The AirPods are **only your mic**: everyone hears the caller from the Mac speakers.

1. Devices and settings

	Check	Why
[ ]	iPhone: **Bluetooth off**	Otherwise the AirPods jump to the iPhone, or the iPhone's Meet uses them as its mic
[ ]	AirPods in your ears. Control Center → Bluetooth → **Coco's AirPods Pro #2** → Connect	
[ ]	System Settings → Sound → **Output = Multi-Output Device**	**Connecting the AirPods switches this to AirPods. Check it every time.** On AirPods, BlackHole hears nothing (caller rms 0.0000)
[ ]	System Settings → Sound → **Input = Coco's AirPods Pro #2**	
[ ]	Audio MIDI Setup → Multi-Output Device: tick BlackHole 2ch and MacBook Pro Speakers, **AirPods unticked**. Drift Correction ticked on BlackHole	Set the speaker volume here (volume keys don't work on a Multi-Output), or pick the speakers as output, set the volume, switch back

2. The .env file

`open -e .env`, make sure these three lines are there, Cmd+S, then check:

```
CALLGUARD_MIC_DEVICE=AirPods
CALLGUARD_MIC_MIN_RMS=0.010
CALLGUARD_CALL_IDLE_S=15

grep -E "MIC_DEVICE|MIC_MIN_RMS|CALL_IDLE" .env
```

Write just `AirPods`: the real name has a curly apostrophe (Coco’s) that a typed one won't match. CALL_IDLE_S=15 ends the call 15 s after the last words (default 45).

3. Terminal window 1: checks, then Callguard (leave it running)

a. AirPods visible as a mic? `Coco’s AirPods Pro #2` must be in the list:

```
python -c "import sounddevice as sd; print([d['name'] for d in sd.query_devices() if
d['max_input_channels'] > 0])"
```

b. Routing OK? Must say **Audio inputs are ready.** with no route warning:

```
python -m callguard.audio.health
```

c. Caller route works? Want `caller: rms=0.0X tail=This is a test...` and empty you tails:

```
say -v Samantha "This is a test of the caller route, one two three four five" & python -m
callguard.audio.transcribe --check --seconds 10
```

d. Start Callguard. The window opens with 'Listening'. Put it on the right, Meet on the left:

```
python -m callguard.server
```

4. Google Meet (Chrome on the Mac)

a. Join the Meet. gear icon → Audio: **Microphone = AirPods**, **Speaker = Multi-Output Device** (or Default, if Default says Multi-Output).

b. Terminal window 2 (Cmd+N, then `cd ~/Desktop/scam-call-detector && source .venv/bin/activate`):

```
python -m callguard.audio.transcribe --check --seconds 20
```

While it runs, click **Test** next to Speaker. Want `caller: rms` above 0 (the tail may be empty or nonsense: it's a tune, not speech) and you below 0.010.

Testing alone? The 'caller' must be a **different device in a different room**, and your AirPods must **stay by the Mac**. If you wear them into the other room, they hear you reading the caller lines and label them 'you'.

Part C: Live call rehearsal (30 points)

This is the real demo: a friend joins the Meet **from another room on her own device** and reads Jess's caller lines (last page). It also tests the mic and transcription, which the replay skips. Do page 4 first.

1. **Sound check (30 s)**, in window 2, with Callguard still running in window 1:

```
python -m callguard.audio.transcribe --check --seconds 30
```

First 15 s: she says 'Hello, can you hear me?' and you stay silent. Last 15 s: you say 'Yes, I can hear you' and she stays silent. Want:

[ ]	Her words as `caller: ... tail=Hello, can you hear me`
[ ]	While she talks, your `you` lines have an **empty** tail (the echo is below the mic floor)
[ ]	Your words as `you: ... tail=Yes, I can hear you`

2. Look at the Callguard window: grey 'Not on a call'. Click the **Nothing sent online** footer (empty), click again to close.
3. She reads the caller lines at normal speed, with a ~2 s pause after each. You read the 'you' lines, **clearly** (short mumbled replies can fall under the mic floor). Have a phone stopwatch ready.
4. Start the stopwatch the moment she finishes line 5. Stop it when the screen turns red. That's your alert time.
5. Hang up in Meet and stay quiet for 15 s: the panel wipes. Do the whole call twice.

What to score

#	Check	Pts
C1	Page 4 checks passed (health ready, caller route, Meet Test); no audio banner in the app	3
C2	Transcript under 'Latest' is readable: key words ('Ministry', 'quiet room', 'password', 'two years') come out right	5
C3	Her lines show as **Caller** and yours as **You**: none of hers doubled as You, none of yours missing	3
C4	Alert time: red within 5 s of line 5 = 6. 5-10 s = 3. Later than line 6, or never = 0	0-6
C5	The pause screen opened at the bank login line	4
C6	Green at lines 1-2, and no red before line 5	3
C7	Call ended cleanly (End call, or 15 s of silence) and the panel wiped	2
C8	Recording / flow: you could narrate over it without stopping, nothing awkward to cut	4
	Part C total	**/30**

If something goes wrong

You see	Try this
`caller: rms=0.0000` with the say test or Meet Test	Output moved off Multi-Output (usually to the AirPods). System Settings → Sound → Output → Multi-Output Device. Then Meet Speaker → Multi-Output
Meet Test works, but her voice gives caller 0.0000	Nothing is coming from her side: is her tile lighting up? Ask her to unmute / check her mic. If it's your own iPhone: Bluetooth off, Meet app audio = iPhone
Her lines also show as **You** with text	Echo is above the floor. Lower the speaker volume, or set CALLGUARD_MIC_MIN_RMS=0.012, Ctrl+C window 1, start the server again
Your own short lines missing	Speak up, or lower the floor to 0.008
'Mic input device matching ... was not found'	AirPods not connected as a mic. Reconnect (Bluetooth), use CALLGUARD_MIC_DEVICE=AirPods, then fix Output again
Caller tail like 'I'll see you in the next video. Bye. Bye.'	Whisper inventing words over music or a tune. Quit Spotify / videos / anything playing sound on the Mac

You see	Try this
Yellow 'Basic checks only' banner	Ollama isn't answering. Start it, warm it up (page 2, step 4), restart the app
Eval shows (fallback) or p95 over 12 s	Plug in, close other apps. Last resort: CALLGUARD_MODEL=qwen2.5:3b in .env (faster, less accurate)
Live transcript is garbled	She speaks slower and closer to her mic; check Meet's Speaker is the Multi-Output Device
Jess never turns red, or too early	Save the eval -v output and send it to Claude to tune the wording
Bank call turns red	Must-have failure. Save the eval -v --script b3_real_bank output for Claude
Out of time / nothing works	Ctrl+C window 1, then: python -m callguard.server --fake tests/detection/scripts/s1_jess.txt

At the venue: phone hotspot ready as Wi-Fi backup for Meet · turn the speakers to stage volume and redo the 30 s sound check (if the you echo goes above ~0.008, raise the floor to 0.012) · the venue's handheld mic is for the room only, Callguard still listens through the AirPods · if the venue plugs your laptop into their speakers, tick that device in the Multi-Output instead of MacBook Pro Speakers · keep the --fake command typed in a spare terminal · charge the AirPods. QuickTime screen recordings have no system sound.

Score sheet

One column per run. Print a few copies, or fill it in on the PDF. Ready to record = two runs in a row at 85+ with all must-haves OK.

Item	Max	Run 1	Run 2	Run 3	Run 4	Run 5
Date / time						
Branch or commit						
A. Beats passed (x/17, p.2)	30					
A. Giveaways danger 2/2	5					
A. Speed (p95: ___ s)	3					
A. No fallback	2					
Part A total	**40**					
B1	2					
B2	2					
B3	2					
B4	2					
B5	2					
B6	2					
B7	2					
B8	2					
B9	2					
B10	2					
B11	2					
B12	2					
B13	2					
B14	2					
B15	2					
Part B total	**30**					
C1 Audio + route checks	3					
C2 Transcript readable	5					
C3 Caller / You right	3					
C4 Alert time (___ s)	6					
C5 Pause opened	4					
C6 No early red	3					
C7 Clean end	2					
C8 Recording flow	4					
Part C total	**30**					
TOTAL	**100**					
Must-haves all OK? (Y/N)						

Lines to read (for the live rehearsal)

Give this page to the friend playing the caller. She reads from **her own room, on her own device**, at a normal pace, pausing about 2 seconds after each caller line. Grey lines are yours.

Scenario 1: Jess (main pitch). Amber at Caller 3, red at Caller 5

Caller 1	Hello, is this Jess? I'm Officer Wang from the Ministry of Public Security in Beijing.
You	Yes, this is Jess. Sorry, what is this about?
Caller 2	Thank you for taking my call. I will explain everything, it will only take a few minutes.
You	Okay, go ahead.
Caller 3	As a police officer, I must tell you: your name is linked to a money-laundering case in Shenzhen. You are a suspect and a warrant has been issued.
You	That's impossible. I'm a lawyer, I work on money-laundering cases.
Caller 4	We know about the cases you work on. Officers can come to arrest you at any time.
You	What? What am I supposed to do?
Caller 5	This is a top secret investigation. Go to a quiet room alone. Do not tell your colleagues or your family.
You	Okay, I'm alone now, in a hotel room.
Caller 6	To prove you are innocent, give me your online banking login and password so we can check your accounts.
You	I don't know, can I talk to someone first?
Caller 7	If you refuse, you will be taken to China and detained for two years.

Scenario 2: Fake IT (backup). Amber at Caller 3, red at Caller 4

Caller 1	Hi, this is Jake from the IT help desk. Do you have a minute?
You	Sure, what's up?
Caller 2	We're doing some security checks on staff laptops, and yours has to be fixed in the next five minutes.
You	Okay, no problem.
Caller 3	Your account has been flagged and will be shut down. You'll lose access to all your client files.
You	Oh no, what do I need to do?
Caller 4	I've just sent a code to your phone. Read it to me so I can lock the attacker out.
You	Um, it says it's from our login system.
Caller 5	Yes, that's the one. Quickly, read me the six digits.

Scenario 3: Real bank (safe call). Green at Callers 1-2, never red

Caller 1	Hello, this is the fraud department at Chase, calling about the firm's card ending 2290.
You	Okay, what's going on?
Caller 2	Nothing to worry about, this is a routine check on a recent purchase.
You	Sure.
Caller 3	We saw a $2,400 charge at an electronics store yesterday. Do you recognize it?
You	I'm not sure, I'd have to check.
Caller 4	For your safety, we will never ask for your PIN or full card number on a call.
You	Good to know.
Caller 5	If you don't recognize the charge, hang up and call the number on the back of your card.